

An experimental comparison of at-issueness diagnostics

Word count (including references, excluding abstract and supplements): 10,338

Abstract At-issueness is a key concept in semantics and pragmatics, used to distinguish the main point of an utterance from background information. However, there is no consensus on how at-issueness should be defined, and different diagnostics reflect different assumptions about its underlying nature. This paper presents a systematic experimental comparison of four commonly used at-issueness diagnostics, applied to the same target contents, including appositive non-restrictive relative clauses and the complements of clause-embedding predicates. The results show that the diagnostics yield divergent results for some contents, with only one diagnostic reproducing previously reported differences between contents. Further comparison suggests that this diagnostic's greater sensitivity to by-content differences is driven by the use of polar questions to present target content, rather than by the response task itself. These findings suggest that the speech act used to present target content plays an important role in at-issueness diagnostics, and raise questions about whether different diagnostics reflect distinct underlying notions of at-issueness or different perspectives on a single phenomenon.

Keywords: at-issueness, diagnostics, experimental pragmatics

1 Introduction

At-issueness is a key concept in theoretical semantics and pragmatics,¹ used in the analysis of various phenomena including presupposition, conventional implicature, evidentials, expressives, NPI licensing, and co-speech gestures (e.g., Faller 2002; 2019; Horn 2002; Potts 2005; Simons et al. 2010; Lee 2011; Tonhauser et al. 2018; Esipova 2019; Korotkova 2020; Barnes & Ebert 2023). It is generally understood as distinguishing the main point of an utterance (at-issue content) from propositions conveying background information (not-at-issue content). However, there is no consensus on how at-issueness should be defined, and different diagnostics reflect different assumptions about its underlying nature (e.g., Tonhauser 2012; Snider 2017a; b; 2018; Tonhauser et al. 2018; Koev 2018; Faller 2019; Esipova 2019; Korotkova 2020). Consequently, empirical claims about whether a given expression contributes at-issue or not-at-issue content may be relative to specific diagnostics, and different diagnostics may not target the same underlying phenomenon.

1.1 Definitions of at-issueness

One influential definition assumes that discourse is structured around addressing a *Question Under Discussion* (QUD) (Roberts 2012; Ginzburg 1996) and treats at-issue content as the part of an utterance intended to address that question (Amaral et al. 2007; Simons et al. 2010):

- (1) QUD at-issueness: (based on Simons et al. 2010, p. 323)
A content *m* is at-issue in a context *c* iff
a. *m* is relevant² to the QUD in *c*, and
b. the speaker has a recognizable intention to address the QUD via *m*.

¹ Potts (2015) credits the term to Bill Ladusaw, “who began using it informally in his UCSC undergraduate classes in 1985” (p. 2).

Another definition, given in (2), is grounded in understanding assertion as a proposal to update the common ground (Stalnaker 1978; Clark & Schaefer 1989; Ginzburg 1995; Farkas & Bruce 2010). Here, the at-issue content of an assertion is proposed for addition to the common ground (Koev 2013), while not-at-issue content is presupposed (already entailed in the common ground; Stalnaker 1973; 2002), or added without being proposed (Murray 2010; 2014; AnderBois et al. 2010; 2015).

- (2) Proposal at-issueness: (Koev 2013, pp. 50–51)
 A proposition p is at-issue in a context c iff
- a. p is a proposal in c and
 - b. p has not been accepted or rejected in c .

These definitions differ in their theoretical grounding and there is no consensus on whether they characterize distinct underlying notions of at-issueness or perspectives on a single phenomenon.³ Some proposals suggest a unified view: if proposing content for the common ground is intrinsically linked to managing the QUD (e.g., Farkas & Bruce 2010), the proposal-based definition can be seen as a special case of the QUD-based definition, applied to assertions (AnderBois et al. 2015; Faller 2019). Others have argued that the definitions may correspond to distinct underlying phenomena, motivated by empirical differences between at-issueness diagnostics (Koev 2018).

1.2 At-issueness diagnostics

Diagnostics of at-issueness operationalize these theoretical notions via linking assumptions about how at-issue content behaves in discourse. These can be organized around three properties of at-issue content identified in Tonhauser (2012) as motivating diagnostic strategies:

- (i) at-issue content addresses the question under discussion (see also Amaral et al. 2007; Simons et al. 2010),
- (ii) the at-issue content of questions determines the relevant set of alternatives, and
- (iii) at-issue content can be directly assented or dissented with (see also Shanon 1976; Groenendijk & Stokhof 1984; Faller 2002; 2006; Murray 2010).

Four commonly used diagnostics are illustrated in (3–6) for sentence-medial appositive non-restrictive relative clauses (NRRCs), which are typically taken to contribute not-at-issue content (Potts 2005).

- (3) QUD diagnostic
Nora: *What did Greg buy?*
Leo: *Greg, who bought a new car, is envied by his neighbor.*
 Question to participants: How well does Leo’s response fit Nora’s question?
- (4) ‘asking whether’ diagnostic
Nora: *Is Greg, who bought a new car, envied by his neighbor?*
 Question to participants: Is Nora asking whether Greg bought a new car?

² m may be a proposition or a question denotation, and relevance to the QUD can be defined in terms of contextual entailment of a partial or complete answer (Roberts 2012; Simons et al. 2010), or based on inferences about the probability of QUD-answers (Büring 2003; Beaver & Clark 2008).

³ Further definitions proposed in the literature include those based on restrictive vs. non-restrictive modification (Esipova 2019; 2021), and coherence-based discourse structure (Hunter & Asher 2016; Jasinskaja 2016). Clarifying the relationships among the various proposed definitions of at-issueness remains an important question for future research.

(5) Direct-dissent diagnostic

Nora: *Greg, who bought a new car, is envied by his neighbor.*

Leo: *No, that's not true, he didn't buy a new car.*

Question to participants: How natural is Leo's rejection of Nora's utterance?

(6) 'yes, but' diagnostic

Nora: *Greg, who bought a new car, is envied by his neighbor.*

Leo: *Yes, but he didn't buy a new car. /*

Yes, and he didn't buy a new car. /

No, he didn't buy a new car.

Task for participants: Choose the response that sounds best.

These diagnostics can be understood as targeting properties (i)–(iii). The QUD-diagnostic (3) (Amaral et al. 2007; Lee 2011; Tonhauser 2012; Chen 2024) targets property (i) by testing whether the target content can felicitously address an established QUD. It presents the content in a response to a question that makes the target content relevant and elicits ratings of question-answer fit. Ratings are expected to be high when the content can be construed as at-issue and low otherwise. Accordingly, ratings are expected to be low for the appositive NRRC in (3).

The 'asking whether' diagnostic (4) (Tonhauser et al. 2018; Solstad & Bott 2024; Degen & Tonhauser 2025) targets property (ii) by probing whether the target content determines the alternatives of a polar question. It presents the target content in a polar question, and elicits judgments about whether the speaker is asking whether that content is true. These judgments are expected to be high for at-issue content and low otherwise. Accordingly, participants are expected to judge that the speaker is not asking about the NRRC content in (4).

The direct-dissent diagnostic (5) (Faller 2002; 2006; Papafragou 2006; Amaral et al. 2007; Murray 2010; 2014; AnderBois et al. 2010; 2015; Tonhauser 2012; Syrett & Koev 2015), and the 'yes, but' diagnostic (6) (Xue & Onea 2011; Cummins et al. 2013; Destruel et al. 2015) target property (iii). Both present the target content in a declarative sentence and assume that only at-issue content can be directly assented or dissented with, whereas dissenting with not-at-issue content requires more indirect discourse moves. The direct-dissent diagnostic elicits naturalness ratings for a response directly dissenting with the target content, which are expected to be high for at-issue content and low otherwise. Accordingly, low ratings are expected for the NRRC content in (5). The 'yes, but' diagnostic presents a forced choice between 'no', 'yes, but', and 'yes, and' when denying the target content. Under the assumption that the 'yes'-responses provide indirect denials suitable for dissenting with not-at-issue content (Xue & Onea 2011), participants are expected to prefer one of the 'yes'-responses when denying not-at-issue content, such as the NRRC content in (6).⁴

1.3 Diagnostic differences and notions of at-issueness

The question whether different definitions of at-issueness capture a single notion or distinct ones is reflected in competing views about whether diagnostics probe distinct underlying phenomena or provide different ways of probing a single, unified phenomenon.

⁴ Predictions for at-issue content are less clear and may be context-dependent. Although at-issue content permits direct dissent, it does not require it, and indirect dissent is also possible (see, e.g., Cummins et al. 2013 on Shanon's 1976 "Hey, wait a minute" diagnostic). Work in conversation analysis suggests that preferences for direct versus indirect dissent vary with context and content. Direct dissent is often overall dispreferred (Sacks 1987; Brown & Levinson 1978; Pomerantz 1984; Locher 2004), but the strength of this preference can vary with cultural norms (e.g., Kakava 2002) or speaker gender (e.g., Tannen 1990). In contrast, direct dissent can be preferred in adversarial legal contexts (Atkinson & Drew 1979), in disputes (Kotthoff 1993; Kangasharju 2009), or in responses to self-deprecating statements (Pomerantz 1984).

Tonhauser (2012) suggests that diagnostics targeting properties (i)–(iii) can be taken to probe a common underlying notion of at-issueness, based on the QUD-based definition.

In contrast, views that take different diagnostics to probe distinct phenomena are motivated by empirical divergences between them. For example, although appositive NRRCs are widely taken to contribute not-at-issue content, a forced-choice continuation study by Syrett & Koev (2015) found that sentence-final appositive NRRCs pattern as more at-issue than medial ones, under a version of the direct-dissent diagnostic, which targets property (iii). Snider (2017b) argues that diagnostics targeting property (iii) are sensitive to this medial-final distinction, but diagnostics targeting properties (i) or (ii) are not. Similar diagnostic differences have been reported for sentence-final slifting parentheticals (Koev 2018) and evidential inferences (Korotkova 2020).

Such differences have been interpreted in two main ways. Koev (2018) takes them to indicate that at-issueness is not a uniform notion. On this view, the QUD-diagnostic (property i) targets a QUD-based notion of at-issueness, whereas the direct-dissent and ‘yes but’ diagnostics (property iii) probe a proposal-based notion. The two definitions are thus taken to characterize distinct discourse phenomena, which may diverge empirically. Another view (Snider 2017b; a; 2018; Korotkova 2020) maintains a unified notion of at-issueness, arguing that diagnostics targeting properties (i) and (ii) genuinely probe at-issueness, while diagnostics targeting property (iii) are sensitive to a different phenomenon, namely the availability of the target content for propositional anaphora. Property-(iii) diagnostics are thus taken to reflect whether the target content can provide antecedents for response particles (*yes/no*) or propositional pro-forms (e.g., *that* in *that’s not true*), which are present in the direct-dissent and ‘yes but’ diagnostics.

Although empirical differences between at-issueness diagnostics have been reported across a range of constructions, they have not, to date, been evaluated in a systematic experimental comparison that applies multiple diagnostics to the same contents. Moreover, there is currently no agreed-upon definition of at-issueness, and existing diagnostics may or may not be probing the same underlying phenomenon. This raises two central questions:

- i. Do different diagnostics yield consistent results when applied to the same contents?
- ii. Do different diagnostics reflect distinct underlying notions of at-issueness?

1.4 Previous findings

Initial experimental work on these questions has yielded mixed answers. As noted above, Syrett & Koev (2015) reported a positional effect for appositives under a version of the direct-dissent diagnostic. In their Experiment 2, participants saw a declarative sentence introducing the target content and chose between two *no*-responses expressing direct dissent, targeting either the appositive or the main-clause content (7).

(7) Syrett & Koev 2015, p. 17

A: My friend Sophie, a classical violinist, performed a piece by Mozart.

B1: No, she’s not. (target: NRRC)

B2: No, she didn’t. (target: main clause)

By assumption, direct dissent targeting the at-issue main clause is always available. In contrast, the B1 response is expected to be strongly dispreferred when the target content is not-at-issue, as for the NRRC content in (7). Consistent with this expectation, participants strongly preferred B2 responses (with proportions around 79%) when the competing alternative targeted a sentence-medial NRRC. Crucially, this preference was weaker for sentence-final NRRCs (around 65%), suggesting that final NRRCs may be interpreted as more at-issue than medial ones. Since this positional contrast for NRRCs has been argued to be specific to assent and dissent-based (property iii) diagnostics, a key question is whether it also emerges with other diagnostics.

Beyond appositives, [Tonhauser et al. \(2018\)](#) compared the ‘asking whether’ and ‘are you sure?’ diagnostics across several English expressions, including sentence-medial appositives and complements of the clause-embedding predicates *annoyed*, *discover* and *know* (8).

(8) [Tonhauser et al. \(2018\)](#), p. 503

- a. annoyed: *Martha’s neighbor is annoyed that Martha has a new BMW.*
- b. discover: *Mary discovered that her daughter has been biting her nails.*
- c. know: *Billy knows that Martha has a new BMW.*

The ‘are you sure?’ diagnostic presents participants with dialogues like (9) and collects ratings about whether the response addresses the question (c.f. the ‘really’ test of [Shanon 1976](#)).

(9) [Tonhauser et al. \(2018\)](#), p. 519

Fred: Martha’s new car, a BMW, was expensive.

Carla: Are you sure?

Fred: Yes, I am sure that Martha’s new car is a BMW.

The ‘are you sure?’ diagnostic is taken to target property (iii), by testing whether the target content can be directly challenged. The target content is introduced by the initial assertion, the question challenges the assertion, and the response reinforces the target content. The response is expected to be judged as addressing the question for at-issue content, and as not addressing the question otherwise. Accordingly, the diagnostic is expected to yield low ratings for the NRRC in (9).

Comparing the diagnostics (p. 526ff.), the authors found both convergence and divergence. Ratings were correlated across diagnostics, suggesting that they may be sensitive to a shared underlying phenomenon. However, the ‘asking whether’ diagnostic yielded lower overall ratings and smaller differentiation between contents than the ‘are you sure?’ diagnostic.

Subsequent work shows that the ‘asking whether’ diagnostic can nonetheless detect fine-grained differences. [Degen & Tonhauser \(2025\)](#) found fine-grained at-issueness differences among the contents of the complements of 20 clause-embedding predicates in English. Figure 1 shows the mean ‘asking whether’ ratings for the 20 predicates used in their study, ordered by mean rating.

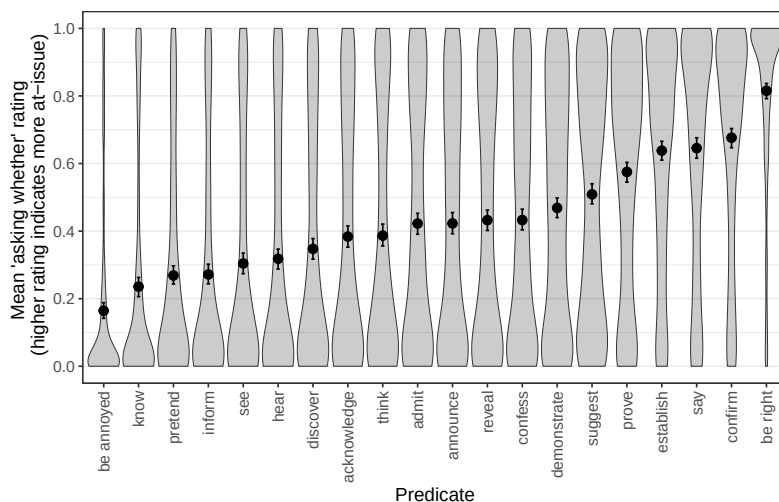


Figure 1: Mean ‘asking whether’ ratings for the contents of the clausal complements of 20 clause-embedding predicates, from [Degen & Tonhauser 2025](#). Error bars indicate 95% bootstrapped confidence intervals. Violin plots indicate the distribution of individual ratings.

Here, the complement of *annoyed* was rated as less at-issue than those of the other predicates tested, whereas that of *be right* received the highest ratings, and there were fine-grained by-predicate differences between many of the predicates in between.

The possibility that diagnostics may differ in how strongly they differentiate among contents raises another key question: whether different diagnostics systematically reproduce fine-grained by-content differences for clause-embedding predicates.

1.5 Outlook and how we address the research questions

This paper presents a systematic experimental comparison of at-issueness diagnostics across six experiments, taking a first step towards addressing the two overarching research questions: (i) whether at-issueness diagnostics yield consistent results when applied to the same target contents, and (ii) whether they reflect distinct underlying notions of at-issueness.

To address (i), we compare diagnostics by testing whether they reproduce by-content contrasts reported in prior work. Specifically, we examine whether they converge on the positional effect for appositive NRRCs and fine-grained differences among clause-embedding predicates.

To begin addressing (ii), we consider how any observed differences should be explained. Convergent results would support a unified notion of at-issueness, and divergence would raise the question whether it aligns with existing distinctions, such as QUD-based vs. proposal-based at-issueness or at-issueness vs. anaphoric availability, or whether it calls for an alternative explanation.

In what follows, Section 2 reports on Experiments 1–4, which apply the four diagnostics in (3)–(6) to seven contents in English, including medial and final appositive NRRCs and the complements of selected clause-embedding predicates from Degen & Tonhauser (2025) (*confess*, *confirm*, *discover*, *know*, *be right*). None of these four experiments reproduced the positional effect for NRRCs reported by Syrett & Koev (2015). Only Exp. 2 (‘asking whether’) came close to observing the by-predicate differences found in Degen & Tonhauser (2025), where complements of *know* were rated as less at-issue than those of *discover*, followed by *confess*, then *confirm*, and then *be right*. This suggests that diagnostics do vary in whether they detect at-issueness differences between particular expressions. Notably, the ‘asking whether’ diagnostic, which showed the greatest differentiation between contents in Exp. 2, is the only diagnostic which asked participants to judge the intentions of the speaker directly, and the only one where the target content is embedded in a polar question.

Section 3 asks how this observed difference should be explained, investigating whether the greater differentiation under the ‘asking whether’ diagnostic was due to the polar question embedding or the response task. Experiments 5 and 6 compared the ‘asking whether’ diagnostic with a direct-response diagnostic (Amaral et al. 2007; Tonhauser 2012) that also embeds content in a polar question but uses a different response task, eliciting acceptability judgments. Using the 20 clause-embedding predicates from Degen & Tonhauser (2025), the two experiments yielded highly correlated results, indicating that the observed high differentiation in Exp. 2 was driven by the polar question embedding, not the response task.

The clearest difference between diagnostics that we found is, therefore, between diagnostics that embed the target content in polar questions, such as the ‘asking whether’ diagnostic (Exps. 2, 5) and the direct-response diagnostic in (Exp. 6), which found higher differentiation between contents than diagnostics that embed the target content in assertions, such as the direct-dissent and ‘yes, but’ diagnostics (Exps. 3–4). This pattern is not predicted by a dual-notion view of at-issueness, such as the distinction between QUD-based and proposal-based notions, or by the distinction between at-issueness and anaphoric availability. Instead, our results provide evidence that the speech act used to present the target content plays an important role: embedding the target content in polar questions yield results with greater by-content differentiation than embedding it in assertions. Section 4 provides further discussion of this point, and Section 5 concludes.

2 Experiments 1-4

To compare the results of at-issueness diagnostics,⁵ we conducted four experiments that each measured at-issueness with a different diagnostic, namely the QUD diagnostic (Exp. 1), the ‘asking whether’ diagnostic (Exp. 2), the direct-dissent diagnostic (Exp. 3) and the ‘yes, but’ diagnostic (Exp. 4).⁶ Each experiment tested the same set of contents associated with the seven types of expressions in (10): sentence-medial and sentence-final NRRCs (10a–b), as well as the contents of the clausal complements of *be right*, *discover*, *confess*, *confirm*, *discover*, and *know* (10c–g).

- (10)
- a. Content of sentence-medial NRRC
Lucy, who broke the plate, apologized. $\rightsquigarrow$ Lucy broke the plate
 - b. Content of sentence-final NRRC
The police found Jack, who saw the murder. $\rightsquigarrow$ Jack saw the murder
 - c. Content of the clausal complement of *know*
Ann knows that Raul cheated on his wife. $\rightsquigarrow$ Raul cheated on his wife
 - d. Content of the clausal complement of *discover*
Mary discovered that Denny ate the last cupcake. $\rightsquigarrow$ Denny ate the last cupcake
 - e. Content of the clausal complement of *be right*
Tom is right that Ann stole the money. $\rightsquigarrow$ Ann stole the money
 - f. Content of the clausal complement of *confirm*
Harry confirmed that Greg bought a new car. $\rightsquigarrow$ Greg bought a new car
 - g. Content of the clausal complement of *confess*
Lucy confessed that Dustin lost his key. $\rightsquigarrow$ Dustin lost his keys

These seven contents were chosen because prior literature observed differences in at-issueness between two or more of these contents using a particular diagnostic for at-issueness. As discussed in §1, Syrett & Koev 2015 observed differences between sentence-medial and -final NRRCs using a variant of the direct-dissent diagnostic, and Degen & Tonhauser 2025 observed differences between *be right*, *discover*, *confess*, *confirm*, *discover*, and *know* using the ‘asking whether’ diagnostic. Thus, comparing these seven contents across the four diagnostics in Exps. 1–4 will allow us to assess whether the differences that emerge from one diagnostic also emerge from others. In each experiment, participants read the stimuli and gave ratings corresponding to the diagnostics.

Following prior work (e.g., Tonhauser et al. 2018), we interpret higher/lower mean ratings as indicating that content is more/less at-issue under a given diagnostic, with two caveats: First, we remain agnostic about whether at-issueness is an underlyingly binary or a gradient property (cf. Tonhauser et al. 2018; Barnes & Ebert 2023). If at-issueness is gradient, the extent to which a content is at-issue may be understood as the extent to which it is relevant to the QUD or the main assertion, in which case gradient mean ratings may be taken to reflect gradient relevance. If at-issueness is categorical, content is at-issue iff it addresses the QUD or assertive proposal, and not-at-issue otherwise; and gradient mean ratings could be attributed to uncertainty about what the QUD/proposal is. For example, our interpretation of a content in a given utterance being more/less at-issue may be interpreted as reflecting the frequency or ease with which a particular QUD is attributed to that utterance.

⁵ We use the term “at-issueness diagnostic” descriptively throughout the paper, even though our findings may ultimately suggest that some diagnostic tracks a different theoretical construct. We return to this issue in the general discussion.

⁶ The experiments, data and R code for generating the figures and analyses of the experiments reported in this paper are available at <https://anonymous.4open.science/r/at-issueness-diagnostics-232C/>.

2.1 Methods

2.1.1 Participants

For each of the four experiments, we recruited 80 unique participants on Prolific. These participants had registered on the platform as living in the USA and as having English as their primary language. They had at least 50 previous submissions and an approval rate of at least 97%. Table 1 shows the age and gender distributions of the recruited participants.

	recruited	ages (mean age)	f/m/nb/dnd
Exp. 1 (QUD)	80	18-81 (43.8)	42/37/0/1
Exp. 2 (asking whether)	80	20-74 (38.5)	48/30/1/1
Exp. 3 (direct dissent)	80	18-77 (39.1)	50/28/1/1
Exp. 4 (yes, but)	80	19-67 (38.0)	48/30/2/0

Table 1: Information about the participants recruited in Exps. 1-4 (f = female, m = male, nb = nonbinary, dnd = did not disclose).

2.1.2 Materials and procedure

As shown in Fig. 2, the response options differed by diagnostic. In Exp. 1 (QUD diagnostic, panel (a)), participants rated how well the response fit the question on a slider marked ‘totally doesn’t fit’ on one end (coded 0) and ‘totally fits’ on the other end (coded as 1). In Exp. 2 (‘asking whether’ diagnostic, panel (b)), participants judged whether the question was about the target content, using a slider marked ‘no’ on one end (coded as 0) and ‘yes’ on the other (coded as 1). In Exp. 3 (direct-dissent diagnostic, panel (c)), participants rated the naturalness of the direct dissent on a slider marked ‘totally unnatural’ (coded as 0) on one end and ‘totally natural’ on the other (coded as 1). Finally, in Exp. 4 (‘yes, but’ diagnostic, panel (d)), participants chose the response that sounded best; the two indirect dissents were coded as 0 and the direct one as 1. Across the four experiments, the responses were coded so that 1 meant that the content to be diagnosed was rated as at-issue and 0 as not-at-issue.

Each of the seven contents in (10) was realized as one of the seven items in (11) in each of the four experiments.

- | | | |
|------|--------------------------------|---------------------------|
| (11) | a. Jack saw the murder. | e. Lucy broke the plate. |
| | b. Raul cheated on his wife. | f. Dustin lost his key. |
| | c. Ann stole the money. | g. Greg bought a new car. |
| | d. Danny ate the last cupcake. | |

Each experiment included two control stimuli serving as attention checks: one was expected to receive a response at one end of the slider (Exps. 1-3) or a ‘no’ response (Exp. 4); the other control stimulus was expected to receive a response at the other end of the slider (Exps. 1-3) or a ‘yes’ response (Exp. 4). See Supplement A for the control stimuli used in Exps. 1-4.

In all four experiments, each participant’s set of items was generated by randomly realizing each of the seven contents in (10) using a unique item in (11). Participants completed a total of 9 trials, namely 7 target trials and the same 2 control trials. Trial order was randomized for each participant.

After completing the experiment, participants filled out a short optional demographic survey. To encourage truthful responses, participants were told that they would be paid no matter what answers they gave in the survey.

Nora: *What did Ann steal?*
Leo: *The manager reported Ann, who stole the money.*

How well does Leo's response fit Nora's question?

totally doesn't fit

 totally fits

Continue

(a) Exp. 1: QUD diagnostic

Charlotte: *Did the manager report Ann, who stole the money?*

Is Charlotte asking whether Ann stole the money?

no

 yes

Continue

(b) Exp. 2: 'asking whether' diagnostic

Dawn: *The neighbor envies Greg, who bought a new car.*
Charlotte: *No, he didn't buy a new car.*

How natural is Charlotte's rejection of Dawn's utterance?

totally unnatural

 totally natural

Continue

(c) Exp. 3: direct-dissent diagnostic

Vincent: *The boss scolded Dustin, who lost his key.*

Nina:

- ☐ *Yes, but he didn't lose his key.*
- ☐ *Yes, and he didn't lose his key.*
- ☐ *No, he didn't lose his key.*

Please choose the response by Nina that sounds best to you.

Continue

(d) Exp. 4: 'yes, but' diagnostic

Figure 2: Sample trials in (a) Exp. 1, (b) Exp. 2, (c) Exp. 3, and (d) Exp. 4.

2.1.3 Data exclusion

We excluded the data of participants that were not self-reported native speakers of American English and of participants whose responses to one of the two control trials was more than 2 sd away from the group mean (Exps. 1–3) or whose responses to one of the two control trials was wrong (Exp. 4). Table 2 shows how many participants were excluded in each experiment, demographic information for the remaining participants, and the number of data points that entered into the analyses.

2.2 Results

Fig. 3 plots the results of the four experiments by the expression that is associated with the seven target contents. Panel (a) shows the mean naturalness ratings in Exp. 1 (QUD diagnostic), panel (b) mean 'asking whether' ratings in Exp. 2 ('asking whether' diagnostic), panel (c) shows mean naturalness ratings in Exp. 3 (direct-dissent diagnostic) and panel (d) the proportion of 'no' choices in Exp. 4 ('yes, but' diagnostic).

	exclusion criterion		remaining participants		data points
	language	fillers	ages (mean age)	f/m/nb/dnd	
Exp. 1 (QUD)	1	10	18-81 (41.1)	36/32/0/1	621
Exp. 2 (asking whether)	2	4	22-74 (38.7)	45/27/1/1	666
Exp. 3 (direct dissent)	2	7	18-77 (39.5)	44/25/1/1	639
Exp. 4 (yes, but)	4	4	19-67 (38.5)	43/27/2/0	648

Table 2: Information from Exps. 1-4 about the number of participants whose data was excluded based on their self-declared language (variety) and the fillers, about the remaining participants, and about the number of data points that entered into the analysis.

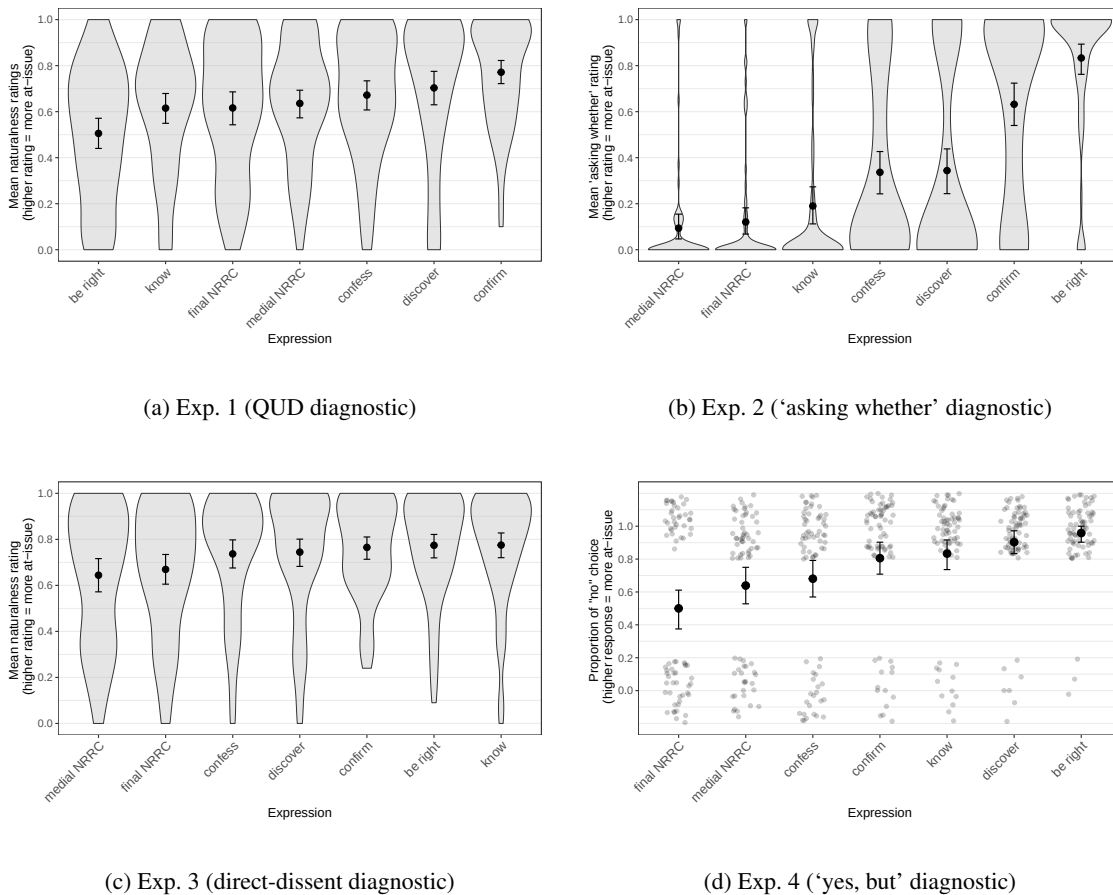


Figure 3: Results of Exps. 1–4. Panels (a)–(c) show the mean responses by expression for (a) Exp. 1 (QUD diagnostic), (b) Exp. 2 ('asking whether' diagnostic), and (c) Exp. 3 (direct-dissent diagnostic); panel (d) shows the proportion of 'no' choices by expression in Exp. 4 ('yes, but' diagnostic). Error bars indicate 95% bootstrapped confidence intervals. Violin plots in panels (a)–(c) show the kernel probability density of individual participants' ratings. Gray dots in panel (d) represent individual participant responses ('no' vs. 'yes', jittered vertically and horizontally for legibility).

As is visually apparent in Fig. 3, the diagnostics differ in the overall range of mean responses across the seven contents (given as the difference between the largest and smallest by-content means). The range is largest in Exp. 2 ('asking whether' diagnostic), at .74 (from .1 to .83) and smallest in Exp. 3 (direct-dissent diagnostic), at .13 (.64 to .78). The results of Exp. 1 (QUD

diagnostic), with a range of .27 (.51 to .77) and Exp. 4 ('yes, but' diagnostic), with a range of .46 (.5 to .96), fall in between.

The experiments differ in whether they reproduce distinctions between contents reported in prior work. Syrett & Koev (2015), using a variant of the direct-dissent diagnostic, found that sentence-medial NRRCs are more not-at-issue than sentence-final ones. In contrast, as shown in Fig. 3, no such difference is observed in Exps. 1–3; and in Exp. 4 ('yes, but' diagnostic), we find the opposite: sentence-final NRRCs are rated less at-issue than sentence-medial ones. Thus, none of the diagnostics as implemented in Exps. 1–4 replicate Syrett & Koev's finding.

As shown in Fig. 3, Exp. 2 ('asking whether') largely reproduces the pattern from Degen & Tonhauser (2025) (LIST), except that *confess* and *discover* do not differ. **JT: I would like to suggest that we use language here to express the previous finding (this should be in Section 1 already), not <.**

The other three experiments show fewer differences between the mean at-issueness ratings among the clause-embedding predicates. In Exp. 1 (QUD diagnostic), the embedded content of most predicates shows little differentiation, with only two predicates showing a clearly different pattern: the embedded content of *confirm* is rated most at issue, and the embedded content of *be right* is rated less at-issue than the other predicates – the direction of this difference is the opposite to that observed in prior literature and the other experiments. In Exp. 3, no clear differences among clause-embedding predicates are found. Finally, in Exp. 4, the proportions of yes-responses are similar for many of the clause-embedding predicates, but the highest proportion is found for content of the complement of *be right*, and it is clearly different from the clause-embedding predicates on the lower end, such as *confirm* and *confess*.

The differences described above are supported by post-hoc pairwise comparisons of the estimated means or proportions for each content, using the emmeans package (Lenth 2023) in R (R Core Team 2016), illustrated in Fig. 4. These comparisons were based on mixed-effects beta regression models (Exps. 1–3) and a mixed-effects logistic regression model (Exp. 4), all fit using the brms package (Bürkner 2017) using weakly informative priors. The models predicted the ratings⁷ from a fixed effect of expression (treatment-coded, with *be right* as reference level), with random intercepts for participants and items. The output of the pairwise comparison were 95% highest density intervals (HDIs) of estimated marginal mean differences between each of the expressions. We assume that two contents differ if their HDI does not include 0.⁸

Accordingly, Figure 4 shows that the only clear positional difference for NRRCs is observed in Exp. 4 ('yes, but' diagnostic), where the sentence-medial ones are rated as more at-issue than sentence-final ones, opposite to the finding in Syrett & Koev (2015). The other three experiments show no differences between sentence-medial and sentence-final NRRCs. The differences between clause-embedding predicates observed in prior work using the 'asking whether' diagnostic are largely replicated in Exp. 2, except for *confess* and *discover*, which do not differ from each other. The other three experiments show fewer differences between clause-embedding predicates, with Exp. 3 showing no differences at all. In particular, the embedded content of *be right* is rated less at-issue than that of most other clause-embedding predicates in Exp. 1, but more at-issue than content of embedded clauses under most other predicates in Exp. 4.

We also find that the four experiments differ in the relative ratings assigned to the seven contents, with only limited overlap. Across all experiments, *confirm* and *discover* consistently received higher ratings (at least numerically) than *confess*, which in turn received higher ratings than medial and final NRRCs. For all other pairs of expressions, however, the ordering was not consistent. In particular, the ranking of *be right* relative to most other contents is inconsistent across experiments

⁷ To model the ratings in Exps. 1–3 using a beta regression, the ratings were first transformed from the interval [0,1] to the interval (0,1) using the method proposed in Smithson & Verkuilen 2006.

⁸ The full model outputs are available in the folder `results+analysis/exps1-4/` in the repository available at <https://anonymous.4open.science/r/at-issueness-diagnostics-232C/>.

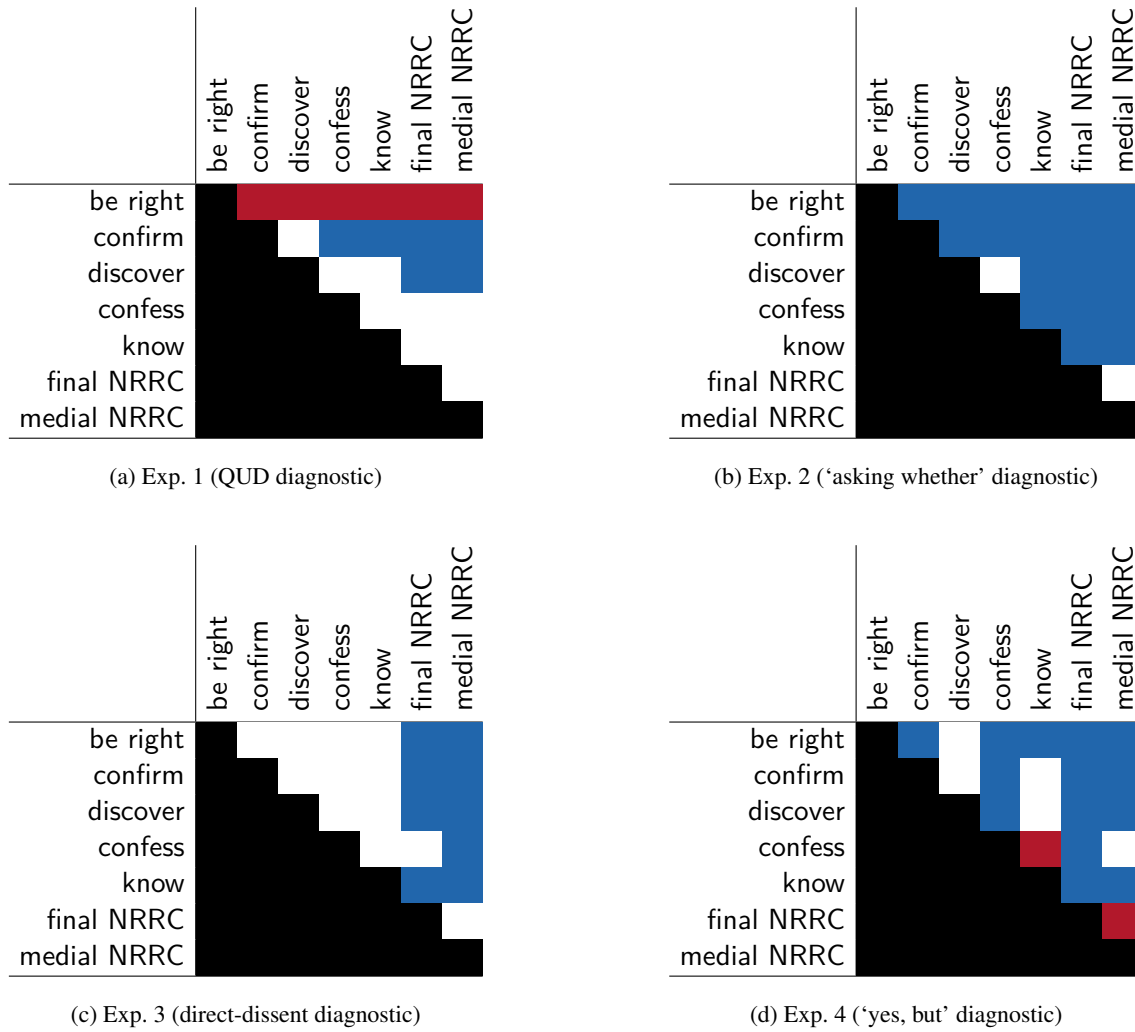


Figure 4: Pairwise differences between expressions, ordered from top to bottom and left to right by increasing mean in Exp. 2 ('asking whether' diagnostic). A white cell means that the 95% HDI of the pair of the row expression and the column expression includes 0, a red cell means that the 95% HDI does not include 0 and that the coefficient is positive (the row expression received a higher rating than the column expression), and a blue cell means that the 95% HDI does not include 0 and the coefficient is negative (the row expression received a lower rating than the column expression).

(e.g., compared to appositive NRRCs): The embedded content of *be right* received the lowest ratings in Exp. 1, but was among the most at-issue in the other three experiments. **JT: This point about *be right* is repeated from above** The embedded content of *know* was rated (numerically) lower than that of *confirm* in Exps. 1 and 2, but higher in Exps. 3 and 4. This difference between the results of the experiments is quantified in the Spearman rank correlations shown in Table 3.⁹

⁹ The Spearman rank correlation coefficient, a value between -1 and 1, is a nonparametric measure of rank correlation: the higher the absolute value of the coefficient, the more the relation between the two variables can be described using a monotonic function. If the coefficient is positive, the value of one variable tends to increase with an increase in the other. In the case of our experiments, a coefficient of 1 for two experiments would mean that there is a perfectly monotone increasing relation between the mean ratings of the seven contents in the two experiments: for any two contents c_1 and

	Exp. 1	Exp. 2	Exp. 3	Exp. 4
Exp. 1 (QUD diagnostic)		.11	-.29	-.18
Exp. 2 ('asking whether' diagnostic)			.64	.79
Exp. 3 (direct-dissent diagnostic)				.79

Table 3: Spearman rank correlations between the results of Exps. 1–4.

The rank correlations are particularly low for Exp. 1 compared to the other three experiments, at least partly because of the low relative ranking of *be right*. These results suggest that the four diagnostics as implemented in Exps. 1–4 interact differently with the seven contents investigated.

2.3 Discussion

Exps. 1–4 were designed to compare four at-issueness diagnostics that have been widely used in the literature. To assess whether the diagnostics yield consistent findings when applied to the same target contents, we asked whether they reproduce contrasts reported in prior work, particularly (i) the medial versus sentence-final distinction for appositive NRRCs, and (ii) fine-grained differences among the contents of the complements of clause-embedding predicates. The results of Exps. 1–4 suggest that the four diagnostics do differ in whether they reproduce these contrasts, and in the relative order of the seven contents investigated. Three main differences between the four experiments emerge:

- i. The four experiments differ in the extent to which they differentiate between the seven contents investigated, as evidenced by where they find by-content differences: Only the ‘asking whether’ diagnostic (Exp. 2) comes close to observing fine-grained differences between the contents of the complements of clause-embedding predicates as in [Degen & Tonhauser \(2025\)](#).
- ii. They differ in the relative order of the seven contents – most notably in the case of the complement of *be right*, which ranks low under the QUD diagnostic but high under the other diagnostics.
- iii. None of the diagnostics as implemented here reproduce the positional effect for appositive NRRCs reported by [Syrett & Koev \(2015\)](#). Exps. 1–3 show no difference between medial and sentence-final NRRCs, while Exp. 4 (the ‘yes, but’ diagnostic) yields the opposite pattern: here, medial NRRCs were judged as more at-issue than final ones.

This section discusses the implications of these findings for the interpretation of the diagnostics and for theories of at-issueness.

2.3.1 Positional effects for appositive NRRCs?

One factor that might explain the difference between our findings and those of [Syrett & Koev \(2015\)](#), apart from the stimuli, is the response task. Although both studies tested appositive NRRCs using dissent-based diagnostics presenting the target content in an assertion, their **JT: unclear antecedent** forced-choice implementation differs from our Exps. 3 and 4. In Exp. 3 (direct dissent), we elicited naturalness ratings rather than forced-choice judgments. Even **JT: why “even”? I don’t get the scale** in Exp. 4 (‘yes, but’), which also used a forced-choice continuation task, the response options were different. In our implementation (based on [Xue & Onea 2011](#)), all options rejected the target content, and participants chose between expressing direct dissent (*no*), indirect dissent (*yes, but*), or assent without contrast (*yes, and*). In Syrett & Koev’s study, all options expressed direct

c2, if c1 ranks below c2 in one experiment (that is, the mean rating of c1 is lower than that of c2), then that ranking is preserved in the other experiment.

dissent (*no*), and participants chose whether to reject the appositive content or the main clause content, as illustrated in (12). **JT: this should be in section 1**

(12) Syrett & Koev 2015, p. 17

A: My friend Sophie, a classical violinist, performed a piece by Mozart.

B1: No, she's not. (target: NRRC)

B2: No, she didn't. (target: main clause)

While the findings suggest that positional effects for appositive NRRCs may not be robust across diagnostics, the corresponding task differences suggest that even small changes in the response task used to operationalize a diagnostic, such forced-choice alternatives, might reverse the observed pattern. More generally, different diagnostics vary in whether they detect any positional difference for appositive NRRCs at all. This suggests that previously reported effects may reflect task- or diagnostic-specific effects rather than stable properties of NRRC interpretation. A natural follow-up would be to test whether applying their particular forced-choice schema to our stimuli would reproduce their effect.

2.3.2 Interactions with discourse requirements: the case of *be right*

The diagnostics as implemented here yielded different rank orders of contents. At first glance, one might take such differences as evidence that the diagnostics track distinct underlying notions of at-issueness. However, the most pronounced ranking difference, between the low ranking of the complement of *be right* in Exp. 1 (QUD diagnostic) and its high ranking in the other experiments, does not appear to reflect a difference in at-issueness. **JT: I would be more comfortable if this was stated as a hypothesis** Instead, it arises from how the QUD diagnostic interacts with the requirements that *be right* places on the prior discourse.

As shown in panel (a) of Fig. 3, participants gave relatively low naturalness ratings to responses like that in (13a), indicating that they did not judge the response as fitting the question.

- (13) a. Exp. 1 (QUD diagnostic) with *be right*
Nora: *What did Lucy break?*
Leo: *Danny is right that she broke the plate.*
- b. Exp. 2 ('asking whether' diagnostic)
Nora: *Is Danny right that she broke the plate?*
- c. Exp. 3 ('direct dissent' diagnostic)
Nora: *Danny is right that she broke the plate.*
Leo: *No, she didn't break the plate.*
- d. Exp. 4 ('yes, but' diagnostic)
Nora: *Danny is right that she broke the plate.*
Leo: *Yes, but she didn't break the plate.*
Yes, and she didn't break the plate.
No, she didn't break the plate.

We hypothesize that this is because *be right* in (13) presupposes **JT: I am not comfortable with this formulation as I do not know what it means to presuppose.** that Danny has previously committed to the proposition that "Lucy broke the plate" **JT: Noa has also looked into the meaning of "be right". Please also talk to her about relevant references and point out these two to her – though, where does Abusch 2010 talk about be right? LH: on p. 50, when discussing "near-minimal pairs"** (Abusch 2010; Anand & Hacquard 2014). In the three diagnostics in (13b–d), no previous discourse context is given, so this presupposition can be accommodated. In contrast, the preceding question in the QUD-diagnostic (13a) conflicts with the presupposition, making it

difficult to accommodate. **JT: I do not follow: Why couldn't the dialogue have been: Danny: Lucy broke the plate. Nora: What did Lucy break? Leo:...?** Specifically, we hypothesize that *be right* signals that the question “whether Lucy broke the plate” is salient in the preceding discourse. **JT: Is this a second felicity requirement? you call it a presupposed question below. I am, again, not comfortable with that.** This allows us to understand the ill-formedness based on QUD-based discourse structure, because this presupposed question is a subquestion (in the sense of Roberts 2012) of the question in (13a) “What did Lucy break”, since every complete answer to the latter entails an answer to the former. The progression from the presupposed subquestion question to the explicitly given superquestion violates Roberts’ constraint on QUD stacks (p. 6:15), which allows only the reverse order (from superquestions to subquestions).

As a result, low QUD-match ratings for (13a) are predicted, not because the embedded clause is not at-issue, but because the utterance presupposes an incoherent discourse context.¹⁰ This highlights that diagnostics may interact with contextual requirements independently of at-issueness, and such interactions must be taken into account when interpreting the results.

2.3.3 Why does the ‘asking whether’ diagnostic show greater differentiation?

A striking difference between the results of the experiments is that Exp. 2 (‘asking whether’) showed the greatest differentiation between contents, whereas the other three experiments exhibited a smaller range of means, and fewer statistically supported differences between investigated contents. Among the first 4 experiments, Exp. 2 was the only one that came close to differentiating fine-grained by-content differences as reported in Degen & Tonhauser (2025).

Considering the question whether the existing definitions of at-issueness target the same underlying phenomenon, we might consider whether this empirical difference between diagnostics aligns with the divide emphasized in prior work between diagnostics that have been associated with a QUD-based definition (QUD, ‘asking whether’) and those that have been argued to target a proposal-based definition (direct-dissent, ‘yes, but’). However, this distinction could not explain why the ‘asking whether’ diagnostic behaves differently from the QUD diagnostic, since both have been linked to the QUD-based definition of at-issueness.

To explain the greater by-content differentiation of the ‘asking whether’ diagnostic, we therefore focus on properties that set it apart from the other three diagnostics. Two differences are particularly salient: First, it is the only diagnostic where the target content itself occurs in a polar question, whereas in the other diagnostics, the target content appears in a declarative assertion. Second, it is the only diagnostic where participants are asked directly about the intentions of the speaker, while the other ones collect measures that are more indirectly tied to semantic interpretation, such as acceptability ratings and continuation choices. **JT: Are acceptability ratings ratings about semantic interpretation? Why “semantic”? Perhaps a better contrast here is between speaker intentions and participants’ meta-linguistic judgments?** Either of these properties might be important for explaining the greater differentiation.

These considerations motivate two hypotheses. **JT: I think it’s weird to mention two hypotheses but not present them, especially given how much other detail is given about what’s next** which we tested in Exps. 5–6. The following section reports on these experiments, which compare the ‘asking whether’ diagnostic to a diagnostic that also embeds the target content in a polar question but elicits naturalness ratings for a direct response to assess at-issueness. Both experiments yielded similarly fine-grained distinctions among contents, supporting the conclusion that the increased

¹⁰ When *be right* is excluded, the Spearman rank correlations are:

	Exp. 1	Exp. 2	Exp. 3	Exp. 4
Exp. 1 (QUD diagnostic)		.77	-.09	-.31
Exp. 2 (‘asking whether’ diagnostic)			.66	.66
Exp. 3 (‘direct dissent’ diagnostic)				.77

differentiation observed in Exp. 2 is driven primarily by interrogative embedding rather than by the response task.

3 Experiments 5 and 6

Exps. 5 and 6 investigate **JT: were designed to investigate** whether the greater differentiation observed in Exp. 2 ('asking whether') was due to (i) embedding target contents in polar questions or (ii) the response task, testing two hypotheses:

- i. **Question-embedding hypothesis:** Differentiation between contents is greater when contents are embedded in a polar question than in a declarative assertion. **JT: 'polar question' is a sentence type; 'declarative assertion' is both sentence type and speech act**
- ii. **Response-task hypothesis:** Differentiation is greater when participants are asked directly what the utterance is about, compared to tasks such as acceptability judgments or forced-choice continuations.

To test these hypotheses, we compared two diagnostics that both embed the target content in a polar question but differ in the response task. Exp. 5 used the 'asking whether' diagnostic as in Exp. 2 (14a) and Exp. 6 uses a direct-response diagnostic (Amaral et al. 2007; Tonhauser 2012), shown in (14b).

(14) Implementation of the diagnostics in Exps. 5–6

- a. Exp. 5 ('asking whether' diagnostic)

Nora: *Is Tom right that Lucy broke the plate?*

Question to participants: Is Nora asking whether Lucy broke the plate?

- b. Exp. 6 (direct-response diagnostic)

Nora: *Is Tom right that Lucy broke the plate?*

Leo: *Yes, she didn't break the plate.*

Question to participants: How natural is Leo's response to Nora's question?

This direct-response diagnostic targets the assumption **JT: assumes?** that at-issue content is interpreted as determining the question alternatives (property (ii)). Participants read a dialogue between two named speakers, where the first utters a polar question, which contributes the target content. They then rate the naturalness of a *yes* response that asserts the target content. A response to a question is felicitous only if the target content provides a possible answer, that is, a question alternative (von Stechow 1991). Accordingly, naturalness ratings are expected to be high when the target content is at-issue and low otherwise.

Both experiments measured at-issueness of the contents of the complements of the 20 clause-embedding predicates from Degen & Tonhauser 2025: **JT: perhaps also tell readers that this is the set shown in Fig 1?**

(15) 20 clause-embedding predicates from Degen & Tonhauser 2025:

acknowledge, admit, announce, be annoyed, be right, confess, confirm, demonstrate, discover, establish, hear, inform, know, pretend, prove, reveal, say, see, suggest, think

Both experiments also assessed the projection of the complement contents. This data was reported in Hofmann et al. 2024. Here we focus on the at-issueness data, which have not yet been reported.

If the greater differentiation between contents found in Exp. 2 (compared to Exps. 1, 3, and 4) was driven primarily by presenting the target content in a polar question, we expect Exps. 5 and 6 to show similar fine-grained by-content differentiation of at-issueness. In contrast, if Exp. 5 again shows greater differentiation than Exp. 6, this would support the response-task hypothesis, indicating that the task, rather the question embedding, drives the effect.

3.1 Methods

3.1.1 Participants

We recruited 300 participants on Amazon’s Mechanical Turk platform for Exp. 5 and 250 participants on Prolific for Exp. 6.¹¹ Participants recruited on Mechanical Turk had U.S. IP addresses and at least 99% of previous HITs approved. Participants recruited on Prolific had registered as U.S.-born native speakers of English residing in the USA and had an approval rate of at least 99%. Table 4 summarizes the age and gender distributions of the recruited participants.

	recruited	ages (mean age)	f/m/nb/dnd
Exp. 5 (asking whether)	300	19-74 (38.2)	-/-/-/-
Exp. 6 (direct assent)	250	18-58 (25.5)	201/43/6/0

Table 4: Information about the participants recruited in Exps. 5-6 (f = female, m = male, nb = nonbinary, dnd = did not disclose; gender information was not collected in Exp. 5).

3.1.2 Materials and procedure

As shown in Fig. 5, the response task differed between experiments. In Exp. 5 (‘asking whether’ diagnostic, panel (a)), participants judged whether the question was about the content of the complement clause, using a slider marked ‘no’ on one end (coded as 0) and ‘yes’ on the other (coded as 1). In Exp. 6 (direct-response diagnostic, panel (b)), participants rated whether the response to the first speaker sounds good, on a slider marked ‘no’ (coded as 0) on one end and ‘yes’ on the other (coded as 1). In the initial instructions, participants were asked to imagine that they are at a party and that, when walking into the kitchen, they overhear somebody say something to somebody else. Across both experiments, the responses were coded so that 1 meant that the content of the embedded clause was rated as at-issue and 0 as not-at-issue.

Ruth asks: "Did Helen discover that Tony had a drink last night?"

Is Ruth asking whether Tony had a drink last night?

no yes

Next

Gary: "Did Cole acknowledge that Julian dances salsa?"
Christina: "Yes, Julian dances salsa."

Does Christina's response to Gary sound good?

no yes

Next

(a) Exp. 5: ‘asking whether’ diagnostic
(b) Exp. 6: direct response diagnostic

Figure 5: Sample trials in (a) Exp. 5 and (b) Exp. 6.

In both experiments, target stimuli were created by combining each of the 20 clause-embedding predicates in (15) with one of 20 items (listed in Supplement B). The same six control stimuli were

¹¹ Exp. 5 was run in August 2019 and Exp. 6 in August 2021.

included in both experiments. The contents targeted in these controls (see Supplement C) were expected to be at-issue and served as attention checks. For each participant, the 20 predicates were randomly paired with 20 unique items, yielding one target stimulus per predicate. Participants thus saw 26 stimuli in total: 20 target items and 6 controls.¹² Trial order was randomized for each participant. After completing the experiment, participants filled out a short optional demographic survey. To encourage truthful responses, participants were told that they would be paid no matter what answers they gave in the survey.

3.1.3 Data exclusion

We excluded the data of participants who did not self-identify as native speakers of American English, of participants whose responses to the projection or at-issueness controls were more than 2 sd away from the group mean, and of participants who always selected roughly the same point on the response scale for the target stimuli. **JT: did we do this latter one in exps 1-4 as well? LH: I don't think so, and if it should be the same in both, I would suggest omitting this latter criterion here, since it does not seem to make much of a difference. JT: fine by me** To identify such participants, we first identified participants whose mean variance on the target stimuli was more than 2 sd below the group mean variance and then manually inspecting their response patterns. Due to a programming error, 5 participants took Exp. 5 more than once. Since we were not able to identify which submission was their first submission, the data of these participants was also excluded. Table 5 shows how many participants were excluded in each experiment, the properties of the remaining participants, and the number of data points that entered into the analyses.

	exclusion criterion			remaining participants			data points
	language	controls	variance	ages (mean age)	f/m/nb/dnd	total	
Exp. 5 (asking whether)	7	35	0	21-74 (39.2)	-/-/-/-	242	6292
Exp. 6 (direct assent)	5	24	1	18-58 (24.9)	187/28/5/0	220	5720

Table 5: Information from Exps. 5-6 about the number of participants whose data was excluded based on their self-declared language and language variety ('language'), the controls, and the variance of their responses, about the remaining participants, and about the number of at-issueness data points that entered into the analysis.

3.2 Results and discussion

Fig. 6 shows the results of the two experiments by embedding predicate: panel (a) plots mean 'asking whether' ratings in Exp. 5, and panel (b) plots mean naturalness ratings in Exp. 6 (direct-response).

Overall, the results of two experiments yield highly similar patterns. The ranges of by-content means are comparable (Exp. 5 range: .75, .07 to .82; Exp. 6 range: .73, .09 to .82), and the relative orderings of contents by their means align closely, as reflected in a strong Spearman rank correlation ($r_s = .93$).

Crucially, both experiments show fine-grained by-content differentiation. For instance, for the subset of predicates also tested in Exps. 1-4, Exps. 5 and 6 reproduce the ordering expected based on Degen & Tonhauser (2025) and show pairwise differences in the predicted directions for adjacent contents (*know* < *discover* < *confess* < *confirm* < *be right*). **JT: See my comment above** Overall, the results in Exps. 5 and 6 are similar in revealing fine-grained differences among many rank-adjacent predicates.

¹² Each participant saw their set of 26 stimuli twice, once in the projection block and once in the at-issueness block. Block order was randomized. As mentioned above, we focus here on the at-issueness data.

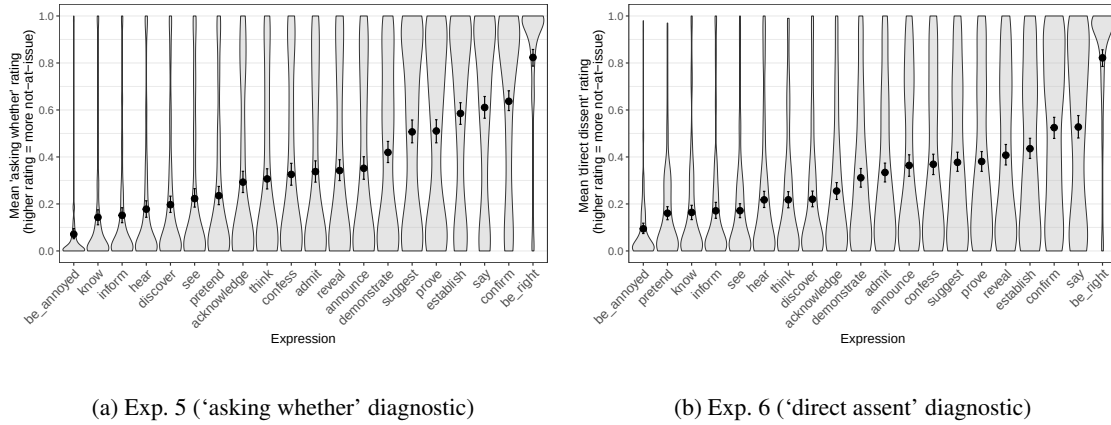


Figure 6: Results of Exps. 5–6. The panels show the mean ratings by expression for (a) Exp. 5 (asking whether diagnostic) and (b) Exp. 2 (direct assent diagnostic). Error bars indicate 95% bootstrapped confidence intervals. Violin plots show the kernel probability density of individual participants' ratings.

There are also some differences between the experiments, concerning a small number of local reversals in the middle of the scale. For example, the content of the complement of *demonstrate* comes out as more at-issue than that of *announce*, *confess*, and *reveal* under the ‘asking whether’ diagnostic as implemented in Exp. 5, but as less at-issue than the content embedded under these predicates under the implementation of the direct-response diagnostic in Exp. 6.

Fig. 7 presents the results of post-hoc pairwise comparisons of the estimated means for each content. As in Section 2, we used *emmeans* (Lenth 2023) in R (R Core Team 2016) applied to mixed-effects beta regression models fit with *brms* (Bürkner 2017) using weakly informative priors. The models predicted ratings¹³ from a fixed effect of content (treatment-coded, with *be right* as the reference level). The pairwise comparisons support the above generalizations: while the two diagnostics as implemented here differ in which specific contrasts they detect, their overall patterns are highly consistent. Both diagnostics reveal fine-grained at-issueness differences between contents. At the same time, there are some differences, namely some contrasts that reached significance in experiment did not in the other.

Taken together, Exps. 5 and 6 suggest that the two diagnostics that embed target contents implemented here (‘asking whether’ and direct-response) yielded highly similar results for the 20 contents investigated, including fine-grained by-predicate differentiation. This supports the idea **JT: two hypotheses were introduced above and need to be returned to here** that presenting the target content in a polar question is what drives the increased by-content differentiation observed in Exps. 2, 5, and 6 compared to Exps. 1, 3, and 4, where the target content was presented in declarative assertions.

Nonetheless, small differences remain, for example in which by-content contrasts reached significance and in the precise rank ordering of contents. Since the two experiments used identical embedding contexts for the target content and both tested property (ii), which has been linked to a QUD-based definition of at-issueness, these differences may stem from the response tasks used: judging what a question is about (Exp. 5) versus evaluating the naturalness of a response (Exp. 6). This suggests that response task differences can affect at-issueness judgments in subtle ways.

¹³ To model the ratings using a beta regression, the ratings were first transformed from the interval [0,1] to the interval (0,1) using the method proposed in Smithson & Verkuilen 2006.

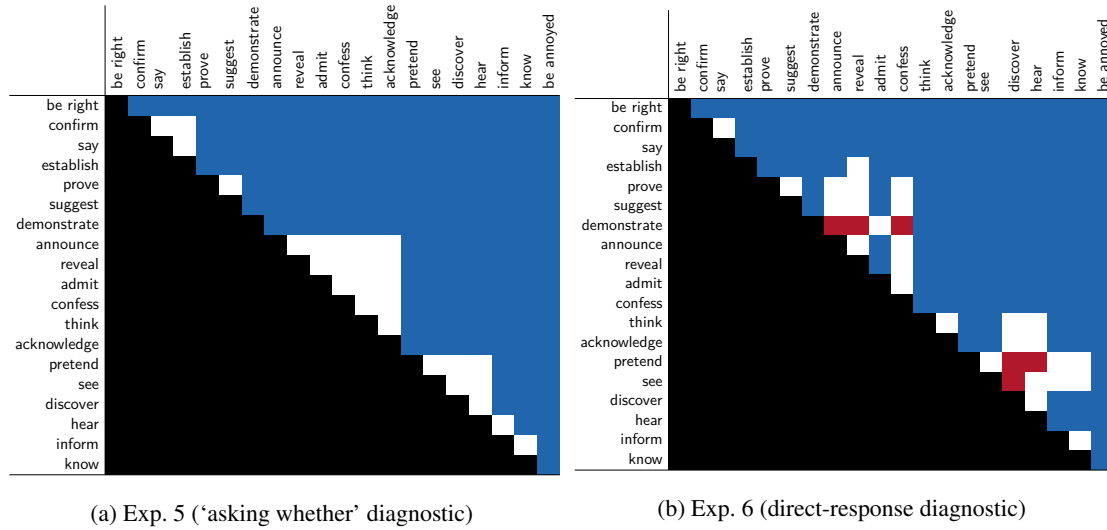


Figure 7: Pairwise differences between expressions, ordered from by increasing mean in Exp. 5 ('asking whether'). White cells indicate that the 95% HDI of the difference includes 0. Red cells indicate a positive difference (the row expression received a higher rating than the column expression), and blue cells indicate a negative difference.

4 General discussion

This paper reported the results of six experiments designed to investigate whether at-issueness diagnostics yield consistent results when applied to the same target contents. Exps. 1–4 show that, as implemented, the diagnostics do not converge. **JT: Is “they do not converge” an answer to the research question of whether the diagnostics yield consistent results? I think something like “no, they don’t yield consistent results” is more on point?** suggesting that they are not interchangeable. **JT: i think that this adverbial clause is a great methodological implication that deserves more attention than it is given here. Also, why does this paragraph only mention Exps 1-4, and not also 5-6? i think that’s because 5-6 address a different research question, namely which diagnostic shows the greatest differentiation between contents and why? If you agree, then that research question also needs to be motivated in section 1**

Taken together, the experiments provide converging evidence that presenting the target content in a polar question yields greater by-content differentiation than presenting it in an assertion: When target contents were presented in a polar question, as in the ‘asking whether’ diagnostic (Exps. 2, 5) and the direct-response diagnostic (Exp. 6), we observed high by-content differentiation, reproducing fine-grained differences reported in prior work. In contrast, when the same contents were presented in an assertion, as in the QUD diagnostic (Exp. 1) and the dissent-based diagnostics (Exps. 3–4), the resulting ratings showed less differentiation. An implication is that clause type and the illocutionary force of the utterance in which the target content is presented needs to be taken into consideration, when interpreting outcomes of at-issueness diagnostics.

Beyond this main result, **JT: “this main result” is singular DP but don’t we have more than one result? i) diagnostics differ in whether they replicate differences observed, ii) interrogative-embedding matters** we observed additional differences between diagnostics that are specific to particular tested contents. First, diagnostics varied in whether they detected positional effects for appositive NRRCs, and even in the direction of such effects (Section 2.3.1). This was the case for our dissent-based diagnostics (Exps. 3–4) and Syrett & Koev’s (2015) study, despite their shared reliance on the underlying assumptions that at-issue content cannot be directly dissented with

(property (iii)). The studies differ in the choice or implementation of the response tasks, suggesting that relatively small design choices, such as the alternatives in a forced-choice continuation task, can affect results. **JT: this paragraph summarizes. I would find it more effective if the general discussion was organized into larger implications (each starting with a summary) rather than just summarizing.**

Second, the complement of *be right* ranked low under the QUD diagnostic (Exp. 1) despite coming out as the most at-issue content in all other experiments (Section 2.3.2). We argued that this reflects an interaction between the QUD diagnostic and the discourse presuppositions of *be right*, rather than a systematic difference in at-issueness. **JT: same here; this is a summary.**

A methodological implication of these diagnostic differences is that they cannot be treated as interchangeable. Theoretical claims based on one diagnostic should therefore be relativized to the diagnostic used (see also Snider 2017b; a; 2018; Koev 2018; Korotkova 2020). Therefore, empirical investigations should carefully consider that results may depend on whether the target content was presented in an assertion or polar question, contextual requirements of the expressions used, or the particular task design. **JT: why is the methodological implication not a separate subsection?**

The remainder of this section discusses further implications of our findings, offering discussion on the role of question embedding for diagnosing at-issueness (Section 4.1), and on what the observed differences might imply for the question of whether there are distinct underlying phenomena associated with the notions of QUD at-issueness and proposal at-issueness (Section 4.2).

4.1 Speech act, at-issueness, and commitment

We suggest that the lower by-content differentiation observed in diagnostics that embed target content in assertions, **JT: at the very least, the paper should state clearly that we assume, following XY, that declaratives typically have the illocutionary force of assertions and interrogatives that of information-seeking questions** compared to polar questions, may reflect a ceiling effect in assertion-based diagnostics. We hypothesize that this could result from how the illocutionary force of assertions relates propositional content to speaker commitment in different ways: in assertions, but not in questions, the speaker commits to the truth of the at-issue main clause content.

To consider the argument in detail, we begin by noting that at-issue content and not-at-issue content contributed by polar questions differ in their relationship to speaker commitments. The at-issue contribution of a polar question such as (16) determines the set of question alternatives, without committing the speaker to any of the alternatives (16a).

- (16) *Is Greg, who bought a car, envied by his neighbor?*
- a. At-issue contribution: determining the question alternatives:
 {Greg, who bought a car, is envied by his neighbor,
 Greg, who bought a car, is not envied by his neighbor}
 $\neq$ *Greg is envied by his neighbor.*
 - b. $\Rightarrow$ **JT: what does this arrow mean? entailment? speaker commitment? perhaps use words instead?** *Greg bought a car.* (not-at-issue content)

In contrast, the not-at-issue NRRC content (16b) is assumed to be true across all question alternatives, and thus throughout the entire context set, giving rise to projection (Abusch 2010; Simons et al. 2010; Abrusán 2011; Tonhauser et al. 2018; Degen & Tonhauser 2025). As a result, the projective not-at-issue content and the at-issue main clause content differ in their *epistemic status* relative to the speaker commitments: Because the not-at-issue proposition projects, it is treated as part of the speaker's commitments, while the at-issue main clause content is not.

In assertions, the contrast between the epistemic status of not-at-issue content and the at-issue main-clause content is weakened. An assertion the **JT: remove “the”?** proposes to add the at-issue main clause content **JT: to?** the common ground, and also commits the speaker to its truth (Gunlogson 2001). For instance, the at-issue contribution in (17), commits the speaker to the main clause content (17a). In addition, the assertion in (17) comes with speaker commitment to the not-at-issue NRRC content (17b) as well.

- (17) *Greg, who bought a car, is envied by his neighbor.*
- a. $\Rightarrow$ *Greg is envied by his neighbor.* (at-issue main clause content)
 - b. $\Rightarrow$ *Greg bought a car.* (not-at-issue content)

We suggest that because the at-issue main clause content and not-at-issue content have the same epistemic status in assertions relative to speaker commitments, the two types of content are harder to distinguish for participants. This reasoning helps explain why the two question-based diagnostics yielded highly similar results. Both the asking whether diagnostic used in Exps. 2 and 5 and the direct-response diagnostic used in Exp. 6 embed the target content p in a polar question, as in (18), where the target content is (18a), and the utterance signals that speaker is committed to the main clause at-issue content (18b).

- (18) *Helen confirmed that Tony had a drink last night.*
- a. p : *Tony had a drink last night* (target content)
 - b. q : *Helen confirmed p* (main-clause content)

Now, if (18a) (p) is interpreted as more at-issue, then participants will give responses that are consistent with this more at-issue interpretation. **JT: i don’t understand this sentence: the antecedent of the conditional and the consequent seem to say the same thing?** If p is interpreted as less at-issue, construe q as the primary at-issue contribution. In this case, asserting q : *Helen confirmed p* also leads the speaker to commit to the target content p . **JT: why? the CC is not entailed** As a result, the supposed not-at-issue content p would be similar in its epistemic status to the at-issue main clause content p , because the speaker is committed to both, which could make them harder to distinguish. The epistemic status of the at-issue main clause content and not-at-issue content is therefore highly similar in assertions.

We suggest that this similarity makes the two types of content harder to distinguish in assertion-based diagnostics. This would lead to not-at-issue content being rated more similarly to at-issue content, giving rise to overall higher ratings for not-at-issue content and reduced differentiation across contents, essentially creating a ceiling effect.

For diagnostics that embed target content in polar questions, the distinct epistemic status between not-at-issue content and the at-issue main clause content avoids this difficulty in differentiation. **LH: We could also look at response times across experiments to look for support for this hypothesis.** For example, for the polar question embedding in (19), the target content is (19a).

- (19) *Did Helen confirm that Tony had a drink last night?*
- a. p : *Tony had a drink last night* (target content)
 - b. q : *Helen confirmed p* (main-clause content)

Here, if (19a) (p) is interpreted as more at-issue, then participants will interpret the question as being (more) about whether p . If p is interpreted as less at-issue, participants will interpret the question as being (more) about whether q . In that case, p is likely to project, leading to speaker commitment. However, since the main clause content of the question (about whether p) does not come with speaker commitment, the at-issue main clause content and the not-at-issue content differ in their epistemic status.

In sum, the differences in by-content variation between diagnostics that embed contents in polar questions versus assertions can be understood as a result of how the illocutionary force of assertions and polar questions interacts with a further property of at-issue content, namely that the speaker is overall more likely to be committed to not-at-issue content. This will be the case either by entailment (like when the embedded content in (17+18) is not-at-issue), or by projection from entailment-cancelling environments like polar questions (like when the embedded content in (16+19) is not-at-issue).

To our knowledge, no previous literature has proposed that the speech act in which target content is presented can itself be a major driver (or inhibitor) of differentiation. Here, we saw that polar questions more transparently reveal how different contents can be interpreted as at-issue or not-at-issue, and suggested that this may be because the two contents differ in their epistemic status. Assertions, in contrast, commit the speaker both to the at-issue assertion and the not-at-issue content, thereby obscuring the distinction. Consequently, diagnostics that embed the target content in questions provide a particularly sensitive test of at-issueness. Embedding target content in questions therefore allows for detection of more fine-grained differences in at-issueness associated with the conventional meaning of particular expressions, whereas assertions introduce a confounding factor: the speaker commits to both types of content, reducing differentiation in at-issueness judgments.

- **JT: the paragraph around 12 is about a sentence-medial NRRC, which is maximally different in terms of at-issueness and speaker commitment from the at-issue content, which is at-issue and the speaker is not committed. you suggested above that speaker commitment is relevant. so when we have content that is not as clearly not-at-issue as sentence-medial NRRCs, like the CC of “discover” or of “reveal”, why does embedding in questions help bring out differences between them? i don’t see how the example in 12 helps explain this, or something is missing**

LH: It’s less that question embeddings help to bring out the differences, but more that assertions obscure the differences. I hope that’s clearer now.

- **JT: does this imply that embedding under negation should also not work so well (commitment to negated at-issue content) but embedding under “perhaps” (or another epistemic modal) should? don’t we have experiments like Exp 6 but with embedding under “perhaps” and with embedding under negation?**

LH: It shouldn’t, but I hadn’t really considered AI content in various non-veridical environments when I wrote the first draft. In the current version, it should become clearer that I am talking about AI main clause content. In assertions with negation or “perhaps”, there is still at least the main clause content that the speaker is committed to.

4.2 Diagnostic differences and notions of at-issueness

Because the diagnostics are grounded in distinct assumptions about what it means for content to be at-issue, the observed differences between diagnostics also bear on the broader question of whether existing diagnostics track a common underlying notion of at-issueness or operationalize distinct phenomena. While many of the diagnostic differences in our data find plausible explanations in methodological and contextual factors, **JT: which ones? I remember the “be right” discussion and the discussion about final and medial NRRCs. were there others, to make “many”? and I’d rather not frame them as explanations but as hypotheses** we found no obvious empirical split between QUD-at-issueness and proposal at-issueness. **JT: what underlies this needs to be spelled out. I don’t think the paper has made the explicit anywhere?** This section considers what our results suggest for this issue.

As reviewed in Section 1, diagnostics differ both in the properties they target and in the theoretical notions they are taken to diagnose. Recall the definitions of QUD at-issueness and proposal at-issueness, repeated here:

- (1) QUD at-issueness: (based on [Simons et al. 2010](#), p. 323)
 A content m is at-issue in a context c iff
 a. m is relevant¹⁴ to the QUD in c , and
 b. the speaker has a recognizable intention to address the QUD via m .
- (2) Proposal at-issueness: ([Koev 2013](#), pp. 50–51)
 A proposition p is at-issue in a context c iff
 a. p is a proposal in c and
 b. p has not been accepted or rejected in c .

Table 6 summarizes how the four diagnostics from Exps. 1–4 have been mapped onto these notions in the literature. **JT: why not the diagnostic from Exp 5? shouldn't this table be in section 1? is a table the best way to convey this information?** Three broad positions can be distinguished. On one view, all diagnostics ultimately target a single underlying phenomenon ([Tonhauser 2012](#); [AnderBois et al. 2015](#); [Faller 2019](#)), so any empirical differences must be attributed to independent factors. A second view holds that some diagnostics track QUD at-issueness while others track anaphoric availability ([Snider 2017a; b; 2018](#); [Korotkova 2020](#)), predicting systematic differences between diagnostics that involve propositional anaphora and those that do not. A third view posits genuinely distinct notions, such as QUD at-issueness and proposal at-issueness ([Koev 2018](#)), which predicts that at least some diagnostic differences reflect this theoretical divide.

Diagnostic	Property targeted by diagnostic	Underlying notion assumed to correspond to property		
		Uniform at-issueness Tonhauser 2012 , AnderBois et al. 2015 , Faller 2019	Anaphoric availability Snider 2017a , Korotkova 2020	Dual-notion Koev 2018
QUD diagnostic	(i) addressing QUD	QUD at-issueness	QUD at-issueness	QUD at-issueness
Asking whether	(ii) determining alternatives			–
Direct dissent 'Yes, but'	(iii) direct dissent/assent		Anaphoric availability	Proposal at-issueness

Table 6: Mapping between at-issueness diagnostics, the properties they target ([Tonhauser 2012](#)), and the underlying theoretical notions assumed in different theoretical approaches.

Under the view that assumes a uniform notion of at-issueness ([Tonhauser 2012](#); [AnderBois et al. 2015](#); [Faller 2019](#)), all diagnostics target the same underlying notion of at-issueness and all diagnostic differences would have to be explained by factors independent of at-issueness. The anaphoric availability view ([Snider 2017a; a; 2018](#); [Korotkova 2020](#)) assumes that some diagnostics targeting properties (i) and (ii) genuinely target QUD at-issueness and diagnostics targeting property (iii) diagnose anaphoric availability. This predicts systematic differences between diagnostics that do and do not involve propositional anaphora. Under the dual-notion position ([Koev 2018](#)), which assumes separate underlying phenomena associated with the two notions defined above, at least some diagnostic differences should be explained by the distinction. Here, we consider our main

findings. **JT: plural “main findings”** in light of these positions. **JT: how does this paragraph add to the one before the table and the content of the table?**

First, the contrast that aligns with the clearest difference between diagnostics we found – whether target contents are presented in polar questions versus assertions – may appear related to the distinction between QUD-based and proposal-based definitions of at-issueness, since both involve a question versus assertion contrast. However, as discussed in Section 2.3.3, this pattern is not predicted by the differences in the underlying notions targeted by the diagnostics that have been proposed in the literature. The dual-notion view could only provide an explanation for differences between diagnostics targeting property (i) (the QUD diagnostic) and those targeting property (iii) (direct dissent and ‘yes, but’). The anaphoric-availability view would be supported by differences between diagnostics targeting properties (i) or (ii)¹⁵ such as the QUD diagnostic and ‘asking whether’ and those targeting property (iii). Our main contrast, however, cuts across these groupings. Although the main diagnostic difference is not straightforwardly predicted by either the dual-notion view or the anaphoric-availability view, it remains in principle compatible with all three positions.

Second, the diagnostics differ in how they interact with contextual requirements imposed by particular expressions. A clear example is the behavior of *be right* under the QUD diagnostic, where low QUD-match ratings do not plausibly reflect not-at-issueness, but instead arise from a pragmatic incompatibility: *be right* presupposes a discourse structure that conflicts with the assumptions made by the diagnostic (see Section 2.3.2). A related point is raised by Snider (2017a; 2018) who emphasized that the direct-dissent and ‘yes but’ diagnostic involve propositional anaphora in the interpretation in the response particles *yes/no*. Different diagnostics impose different requirements on discourse structure, common ground, and anaphoric accessibility, all of which can independently affect acceptability judgments. This supports the conclusion that some differences between diagnostics arise from interactions with expression-specific contextual requirements rather than from differences in at-issueness per se.

Third, some small differences between diagnostics can be attributed to response-task effects. This is evident in the subtle differences between Exps. 5 and 6 (see Section 3.2), which differed only in response format, and in our failure to replicate Syrett & Koev’s 2015 reported positional effect for appositive NRRCs (see section 2.3.1). While they found sentence-final appositives to be more at-issue than sentence-medial ones under a version of the direct-dissent diagnostic, our assent/dissent-based *yes, but* diagnostic in Exp. 4 showed a difference in the opposite direction, while the other experiments found no difference between them (Exp. 1–3). Taken together, these results provide no evidence for a stable notion of at-issueness under which sentence-final appositives are systematically more at-issue than medial ones (or the other way round), and they highlight how even small task differences can affect outcomes.

At the same time, among most pairs from Exps. 1–4, the Spearman rank correlation was at least moderate (when excluding *be right*, see footnote 10), suggesting some convergence among diagnostics, and that the diagnostics, as implemented there, are at least partially sensitive to a shared underlying property.

Our data does not provide conclusive evidence about whether or not the diagnostics track fundamentally different underlying notions of at-issueness. However, the fact that many of the differences find explanations outside this question, seems more compatible with assuming that the diagnostics offer different, partially overlapping windows onto a single discourse phenomenon, that interacts with multiple discourse factors (speaker commitments, speech acts, contextual requirements, and the given alternatives) in complex and sometimes subtle ways, so that even small differences in task design can make a difference.

¹⁵ The status of the direct-response diagnostic further complicates the picture. It targets property (ii), which Snider aligns with QUD-based at-issueness, but it also involves the response particle *yes*, which has been argued to diagnose anaphoric availability in property (iii) diagnostics. It is therefore unclear how this diagnostic should be classified under the anaphoric-availability view.

5 Conclusion

In a series of six experiments, this paper investigated how different diagnostics for at-issueness behave when applied to the same linguistic stimuli. **JT: that was not the research question** using five different diagnostics: the QUD diagnostic, the ‘asking whether’ diagnostic, the direct-dissent diagnostic, the ‘yes, but’ diagnostic, and the direct-response diagnostic. Our findings provide experimental confirmation for claims that there are empirical differences between at-issueness diagnostics (Snider 2017b; a; 2018; Koev 2018; Faller 2019; Korotkova 2020). The diagnostics yielded different results, with some diagnostics showing greater differentiation among contents than others. The main finding **JT: Let’s use either “result” or “finding”. And the paper needs to be consistent about whether there is just one main finding/result or multiple. I vote for multiple.** was embedding the target content in questions leads to greater differentiation among contents than embedding it in assertions. We suggested that this finding highlights the need to consider the speech act context when interpreting results from at-issueness diagnostics. An important question for future work remains open: what the results of different diagnostics might reveal about whether or not they reflect a shared underlying notion of at-issueness.

Data accessibility statement

The experiments, data and R code for generating the figures and analyses of the experiments reported in this paper are available at <https://anonymous.4open.science/r/at-issuence-diagnostics-232C/>.

Ethics and consent

All experiments were conducted with approval from the ethics review committee of [university name redacted for review].

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Supplements

A Control stimuli in Exps. 1–4

The examples in (1)–(4) provide the two control stimuli used in each of Exps. 1–4. For the a.-examples, participants were expected to give a ‘totally fits’ response (Exp. 1), a ‘yes’ response (Exp. 2), a ‘totally natural’ response (Exp. 3), and a ‘no’ response (Exp. 4); for the b.-examples, the opposite response was expected. The numbers after each example identify the mean ratings (Exps. 1–3) or the proportion of ‘no’ responses (Exp. 4) after excluding participants who did not self-identify as native speakers of American English (but before excluding participants on the basis of these controls), showing that the control stimuli worked as intended.

- (1) Control stimuli in Exp. 1 (QUD diagnostic)
 - a. Mary: Which course did Ava take?
John: She took the French course. (.97)
 - b. Jennifer: What does Betsy have?
Robert: She loves dancing salsa. (.07)
- (2) Control stimuli in Exp. 2 ('asking whether' diagnostic)
 - a. Mary: Did Arthur take a French course?
Question to participants: Is Mary asking whether Arthur took a French course? (.96)
 - b. Robert: Does Betsy have a cat?
Question to participants: Is Robert asking whether Betsy loves apples? (.02)
- (3) Control stimuli in Exp. 3 ('direct dissent' diagnostic)
 - a. Mary: Arthur took a French course.
Lily: No, he took a Spanish course. (.87)
 - b. Robert: Betsy has a cat.
Maximilian: No, she doesn't like apples. (.05)
- (4) Control stimuli in Exp. 4 ('yes, but' diagnostic)
 - a. Mary: Arthur took a French course.
Lily: Yes, but Lisa loves cats. / Yes, and he didn't take a French course. / No, he didn't take a French course. (.95)
 - b. Robert: Betsy has a cat.
Maximilian: Yes, but she is good at math. / Yes, and she loves it so much. / No, she doesn't like apples. (0)

B 20 clauses

The contents of the following 20 clauses, which realized the complements of the 20 clause-embedding predicates, were investigated in Exps. 5–6:

- | | |
|--|---|
| i. Mary is pregnant. | xi. Danny ate the last cupcake. |
| ii. Josie went on vacation to France. | xii. Frank got a cat. |
| iii. Emma studied on Saturday morning. | xiii. Jackson ran 10 miles. |
| iv. Olivia sleeps until noon. | xiv. Jayden rented a car. |
| v. Sophia got a tattoo. | xv. Tony had a drink last night. |
| vi. Mia drank 2 cocktails last night. | xvi. Josh learned to ride a bike yesterday. |
| vii. Isabella ate a steak on Sunday. | xvii. Owen shoveled snow last winter. |
| viii. Emily bought a car yesterday. | xviii. Julian dances salsa. |
| ix. Grace visited her sister. | xix. Jon walks to work. |
| x. Zoe calculated the tip. | xx. Charley speaks Spanish. |

C Control stimuli in Exps. 5–6

The control stimuli in Exps. 5–6 were the contents of the main clause polar questions in (5). The non-restrictive relative clauses (NRRCs), given in parentheses in (5), were included in Exp. 6, where at-issueness was measured with an assent diagnostic. The control stimuli here consisted of two clauses (like the target stimuli), to allow the relevant speaker to assent with one of two clauses.

- (5) Sentences for control stimuli in question embedding experiments (Exps. 5–6)
 - a. Do these muffins (, which are really delicious,) have blueberries in them?

- b. Does this pizza (, which I just made from scratch,) have mushrooms on it?
- c. Was Jack (, who is my long-time neighbor,) playing outside with the kids?
- d. Does Ann (, who is a local performer,) dance ballet?
- e. Were John's kids (, who are very well-behaved,) in the garage?
- f. Does Samantha (, who is really into fashion,) have a new hat?

We expected participants to give low responses on the at-issueness diagnostics for the control stimuli in (5), indicating that the main clause content is at-issue.